

ActInf Livestream #030.2 ~ "How to count biological minds: symbiosis, the

Livestream | 1:52:27

hello everybody

welcome to active lab live stream number

30.2 today it is

october 12th 2021

and we're having our second discussion

group discussion

on the paper of matt sims

how to count biological minds symbiosis

the free energy principle and reciprocal

multi-scale integration

welcome to the active inference lab

we are a participatory online lab that

is communicating learning and practicing

applied active inference you can find us

at the links here on this page

this is a recorded and archived live

stream so please provide us with

feedback so we can improve on our work

all backgrounds and perspectives are

welcome here and we'll be following good

video etiquette for live

streams at this short link you can see

some of the past present and future live

streams in the lab

and

in the first tab you'll see the tuesday

numbered live stream so every tuesday we

have a participatory group discussion

two weeks per paper and then there's the

dot zero video which is just sort of a

contextualizing video

and then these other tabs like guest

stream model stream and math stream are

opportunities for just one-off events on

any

topic that people think is relevant for
the lab so get in touch if you want to
join one or co-organize one of these
streams

today in 30.2 we're going to be
continuing to learn and discuss this
awesome paper by matthew sims
how to count biological minds

and

we're just

having fun and discussing

last week we had matt join which was a
great discussion

and

we look forward to everyone's input and

if you're watching in the live chat or

if you're leaving a comment later look

forward to continuing the discussion

today we're pretty much just going to

introduce ourselves and say hello

and then we will

go

wherever we feel relevance

and explore different questions people

can ask about a figure or a section of

the text and then also blue

and others have prepared some discussion

questions to get into

so we can start with the warm-up and

introductions

so i'm daniel and i'm a researcher in

california

and

i think the thing i'm excited about

today

just like with many dot twos

is taking a lot of these cool ideas that

came up in the paper and in the dot zero

and the dot one
and then
synthesizing and consolidating and then
asking okay well now what next
how do we go from this
conversation theory users and resources
reciprocal multi-scale integration
and then apply it to
different systems
so i think that will be a fun topic and
i'll pass to stephen
thank you yes steven here i'm in toronto
on a nice sunny day and um
i'm really interested as well actually
in these building on these ideas i'm
really excited that the folk psychology
and folk biology was brought up and i
think that
that gives permission i think to build
that further in and
maybe extends into
how our phenomenology
connects to our
folk psychology because our folk
psychology allows us all to be
inherent psychologists of ourselves
maybe so that's interesting
and we just enjoy it
analyzing our own reality so i'm going
to pass it over to dave um
yep i'm in the mountains of the
philippines
and uh it's the end of the rainy season
and so we'll have no weather until
next year
um i am pumped up about the
multiple disciplines that are coming
together in the discussion of the paper

that we're doing now and also about
bobby azarian who is not afraid to jump
down into the trenches and face the
lunatics with their machetes and and
even less pleasant uh
arms um
and getting a note together to him to
suggest some topics that we might want
to
talk about
well two two things one off a question
about bobby but first was when you said
there's no weather
it's kind of like what stephen said
about folk psychology
no experience you know just like well
now i'm awake and now experience is
going to stop
it's kind of like weather stopping like
weather is always happening
it's always changing in different ways
and our experience is also always
happening and always in different ways
but just like weather sometimes only
gets studied or discussed in the context
of catastrophe
or pathology
or of extremely unusual events
there's this like baseline weather
that's just the climate
and so
that's very interesting about our
day-to-day
felt realities and then how everybody is
in the driver's seat with respect to
their
experience so sort of the mundaneness of
weather

the mundaneness of experience but also
it's something that can be really
unpacked stephen
yeah exactly the mundaneness is
interesting especially when you think of
the priors that set things up because um
in some ways it's like
the psychologist
in psychology gets to decide
on
the modeling that's going to be used to
look at people whereas in folk
psychology this is the implicit
modeling that we are
framing ourselves around
and of course the same with the folk
biology which this paper brings up and
maybe the folk physics and whatever so
these kind of
implicit priors
um
you can't get away from them and maybe
in a way it brings us into this interest
in are those priors
how much of them are psychological or
are they more like practice based
phenomenological
questions so it's an interesting balance
there cool
and then the second question so bobby um
azarian is going to be doing a guest
stream in november we have a few guest
streams coming up so um
there's multiple but dave what made you
excited about
his presentation or work with the
universal bayesianism
well it's not so much that particular

approach it's that he's been doing a lot
of work on cult formation uh cult
dynamics the growth of violent cults
especially in the us right now and that
directly ties into

philip guerin's

work on the uh neuroscience of delusion

okay very interesting and so i'm just

showing on the screen

those are the dates november 19th

is um with bobby but we'll have um

greg enriquez on the

11th or on october 22nd for number 11

then martin boots for 12 on october 25th

adam saffron on november 8th and then

november 19th so cool guest streams

coming up and of course everyone's

always welcome to join so let's return

yeah i didn't notice you've gotten books

great i am very pleased

yeah

sometimes people just sneak in there you

know

um okay so let's

return to

30.2

and uh

i'm sure blue will join so let's just

save some

tasty pieces for her that she added in

and then otherwise i think we can

maybe let's start with these

questions on

seven

so

blue

in absentee

asynchronous blue

our favorite shade
contributing to the discussion
so
let's see what we
think and if we can connect it to
some of the ideas or sections or figures
of the paper
at any given time can agents act
autonomously as long as there is a
defined prior
and then i think maybe that's connected
to the second question is there a
difference between a cognizer and an
autonomous agent
so how are the terms of
cognition
agent slash agency and autonomy
how do these terms connect
like if we imagine a three-dimensional
space
and we think okay you could have you
know ten zero zero you have ten
cognition zero agency zero autonomy
what is the space of the possible
of cognition agency and autonomy
and then how are these different terms
linked are one out of necessity
uh leading the other or lagging the
other so
stephen and then dave
i mean this question about a defined
prior i mean the other option is
is it if they don't need a defined prior
and they can build up
their
inference
[Music]
process

without having that prior so i i'm not
sure which way that might be the most
important is it is the autonomous agent
obviously we're assuming the agent can
survive

you know this is not a
bolt of lightning or something so it's
something that can have
some sort of non-equilibrium steady
state over time
is it more about the autonomous agent
being able to
act and
process its active inference
independently of others
and that's maybe more key maybe that's
the sort of thing that happens with
cancer as well you know um so i don't
know

welcome blue we're just on slide seven
working through some of your interesting
questions dave

yeah on that the first question um
in the dialogue between christian and
marx solms

they use prior
basically to mean an affect or the other
brain stem resident

um
biological imperatives
and um

i think some would tell you that except
in

anesthesia you know deep deep anesthesia
and perhaps coma there's always
operative priors because you really
can't uh dissociate
the um

homeostatic

goal seeking from the higher processes

since

of the higher processes are

completely dependent on and shaped by

the

uh the homeostatic

uh activities and i wouldn't even accept

coma uh without qualification because of

the i don't know if this research has

been worn out but there are therapists

who have simply proceeded under the

assumption that

um

deeply comatose individuals actually are

taking something away from their

experience and they're finding that in

recovery from coma that yeah if you talk

to people on a daily basis so you talk

about interesting things or unresolved

problems this shapes the way they're

going to

uh talk about their experience

afterwards uh one thing that happens uh

is that people

people's thinking becomes extremely

slow

minutes rather than second

and that if you actually speed up videos

of the response of

comatose individuals

you'll see responsible just like if you

slow down

high resolution videos of infants you'll

see that yes they're highly attuned to

what's going on around them they're

they do have their own priors

that

through which they're responding to
events and actually shaping them in a
healthy diet
interesting
so i think starting with uh
with steven's point and then anyone can
of course can just raise their hand or
write a question live chat
so the question about starting with the
prior
and whether we start with a very defined
prior like a domain specific prior or
whether that's learnt
that is reminiscent of the difference
between first order cybernetics and then
higher order cybernetics like the first
order cybernetics of chemistry is your
priors on chemistry in the second order
would be
your approach to learning
or your dynamics of flexibility around
study habits or around updating your
generative model of chemistry as well as
the second order generative model of
learning
so
as we
start to sort of meet in the middle with
building active inference up
to
describe and model these increasingly
recursive
and symbolic cognizers
will start to see
the movement from models that merely
describe the mountain car
into
models of mental action

into
higher order cybernetic models of mental
action so that will be very
cool
and then um i think you mentioned the
lightning bolt
and so i was coming back to this this um
trilemma or these three terms there's
more terms could be added like
um does lightning bolt have agency does
it have autonomy
well
it sure seems to just act the way that
it's apt to so it seems like an
autonomous system it's not waiting for a
bluetooth connection with something else
so it's high autonomy
and then does it have agency though and
so how is
there a difference
and then cognition especially because
the example of the lightning bolt it's
electricity
so if what we've been talking about with
um professor levin and others is that
like bioelectricity is sort of the
fundamental that's where
biotic and abiotic cognition
come together or where they they blur
so then is a lightning bolt cognitive
and
so that was and then just a last
comment
was the speed of perception and
cognition action
well active inference is about
integrated modeling of perception
cognition and action

so then we're in a quasi-computational framework

there are computationalists among us people who think that cognition is computation as well as people who take more nuanced steps or reject that notion but this is kind of like our clock speed this is like the speed of that loop what's the tick speed of the generative model and that relates to the thickness of our temporal experience like flickering images at 30 per second are perceived as continuous so we don't know exactly what the clock speed is or whether a clock speed is even the apt metaphor but surely there are stimuli that are so close spatially or temporally we cannot tell the difference

so it was just cool to hear like that different ages and different conditions might have different um parameters for their internal generative model and perception action loops blue and then anyone else

so going back to whether or not a lightning bolt um has an agency it certainly seems that like if there is a defined fire like for a lightning bolt there's a defined like prior state that you know like the dielectric constant has to break down for the lightning bolt to be able to occur

so as long as there's that defined prior then

yes like the uh the lightning bolt acts

autonomously
but but
does it have agency is like
well what's the goal of the lightning
bolt
is it to
like minimize the electricity build up
in the cloud or like what would the goal
of a lightning bolt be
um and so
that's kind of what you think about like
even or what i think about
when it comes to
like cells okay the goal of a single
cell organism is to maintain homeostasis
at some you know set point or within
some range
so but a lightning bolt really what's
the goal of a lightning bolt um and so i
think about that
i don't know so it does a lightning bolt
have a goal and and that's kind of where
the sense of agency
comes into play for me um and like yeah
if you define the goal like in a you
know autonomous car like your goal is to
safely transport people from a to b
without running anyone over or getting
smashed like or that's very simple i
know but you know obeying all the
traffic tools and so forth so
you can establish a goal for an
autonomous car but does the car come up
with its own goal
or is that like a human-driven goal and
is the car then a tool
for the humans so so there's like some
fuzzy boundaries here that kind of

always

point toward um the life-mind continuity
thesis which i think we probably haven't
gone to yet but maybe we'll get there
um where life has to

happen at some degree for for
um

cognition to occur maybe or maybe not
very cool so

the goal of the lightning bolt so let's
think of a few you know physical systems
we know and love like a ball rolling
down to the bottom of a bowl
it's minimizing its kinetic energy
differential that's one description not
to say that's the explanation but it's
minimizing something
by rolling to the bottom of the bowl
without fail

similarly when there's an electric
charge buildup between the cloud and the
ground or between two
electric components inside of a
electronic device there's like a zap of
electricity that can also be understood
in terms of minimizing some energy
differential

that's what's interesting about active
inference is we're also thinking about
minimizing
differentials

of

information and so

what stories

are explanations or descriptions of the
ball rolling down kinetically or the
lightning strike electromagnetically
and then are we going to have the exact

same template

of explanation

for

action of biological entities

so that's a and where it is teleology

and directedness

like

is the ball rolling into the bottom of

the bowl is that end directed or is it

just directed only by approximate

teleology like a next step only so

people say evolution has no end it's

just a local optimization process for

example so it's behavior along those

lines um

and then the second point then stephen

and dave was like

is the

system of interest the primary cognizer

or is it being used in an extended or an

instrumental fashion

wasn't sure on the answer there but that

speaks to

matt's paper

with the users and the resources

and so there's cases where you have like

an adaptive active inference agent

utilizing a mere active inferencer

as a tool

like you could be using a thermometer as

a tool as you investigate the cave

but then there's this other

reciprocal case where actually the the

primary cognizers are sort of adaptive

on both sides so there's a lot to

explore there stephen

yeah the idea of things being either

frozen

at a certain state or having the ability
to be a whole to stay in that state
um i think about um well i mean on a big
scale the sun right the sun
has got an energy differential but
um it's it's it's self-maintaining right
because as it keeps going it more more
material can be fused in the middle and
it keeps
so you get into the broader question of
complex adaptive systems of which
non-equilibrium steady-state systems are
a subset so you've got this um
you know there's an interesting point
there and when you have something which
dissipates and can't form a markovian
sort of boundary
um you so it will lose that but there
are times when you can maintain that for
instance
um if you have a globular molten
silica
plunge it into cold water
you can freeze the liquid state it's
still a liquid but it's glass now
so this same
so it's maintaining
that over longer periods of time
um
but is it
is it still not equal is has it has it
formed is it just waiting to allow its
energy gradient to dissipate or is it
now managed to hold its state more
concretely
very interesting has a true
pattern or structural integrity been
reached or are we just sort of

pausing or freeze freeze thaw
on the energy differential just sort of
make the ball made of velcro so it's
still stuck up there
dave yeah there's a hilarious tour de
force by uh stuart bartlett
on
life as a solution to the planet earth's
dilemma of having too much heat and
needing to get rid of it faster and he
breaks down all these amazing processes
culminating with the reverse krebs
citric acid cycle as increasing the
efficiency with which
heat can be radiated by creating just
the right kinds of molecules
in enough concentration to catalyze that
problem of boy it's hot in here
this um it's kind of like the science
fiction classification i don't know what
class you know class one class two class
three civilizations whether their energy
capacity is like less than a star
equal to a star multiple stars

[Music]

some thermodynamic perspectives on
the emergence and the elaboration of
life
when we see
living systems like schrodinger or scott
david as
entropic systems that are resisting
dissipation and you realize that for
every unit of order that is
constituted inside of the markov blanket
that more disorder must occur outside
you know that it can't be um
free lunch and it isn't going to be one

to one either so you're making an excess
of disorder which is what pulls chemical
reactions forward when you order like
for every you know one pound of muscle
organized on your body
more than one pound of muscle
equivalence have to be

um

dissipated

and then it is actually this imperative
that's thermodynamic for life to
continue inventing potentially even
increasingly seemingly inefficient
mechanisms

so

that's sometimes what i

think about when someone talks about the
energy use of cryptocurrency i think you
know if you went into the cell and you
said well we already have the enzyme
making the forward reaction why do we
need the one making the backwards
reaction that's wasting energy

well

it gives flexibility to the cell and it
pushes the reactions forward and it
gives capacity to regulate very
adaptively so it's not always like we
can just look at the um

the

heat producing light bulb and say that
that's the inefficient one so blue and
then stephen

so

i wonder too thermodynamically um
is like life

ultimately causing like the heat death
of the universe right like are we using

up all of the the thermodynamics free energy that's available um with the process of life like we are like in the scott david sense um you know entropy secreting right we're the only entropy i mean life is the only entropy minimizing thing that i can think of but i mean it's just ultimately going to lead to the lack of free energy thermodynamic free energy that's available to do work um which is just an interesting thing to think about and then also i was you know listening to some older live streams and um maybe uh maybe dave actually probably has the knowledge to kind of clarify for me but i always um thought that informational free energy was fundamentally connected to thermodynamic free energy um and so is like in what someone had suggested that um that was not the intention of info like of the free energy principle was never supposed to involve thermodynamics like initially um and i just wonder um is that like some abstract bridge that was formed the connection between infodynamic and thermodynamic um entropy or is that a real concrete um connection and so for for me it's something i've always thought about as as like you know links through physics

and statistical physics like those two things go together

but is there a disconnect there that i'm perhaps not aware of and um was the free energy principle not meant to include thermodynamics so i would love to hear what um you guys think about that yeah dave go for it and then stephen yeah you know i think both entropy and the free energy principle are higher order

metaphors or analogies at least um i'm guessing an entropy is roughly a third level analogy you have to go up to that level of ramification to cover the various cases

um

but there are people working on that i think i sent you a paper by uh sean carroll on the thermodynamic uh interpretation of of what was it the free energy principle i just sent that a couple of days ago i don't know if you saw that i dropped it in

one of the chats

dropped it in the corner of a labyrinth yeah

yeah i'm going to have to go back to buying borges

i think he's i've left too much in his various lab room

stephen

yeah there's a couple of good points there um i suppose one thing in terms of life it's it's trying to stay in that golgi locked zone so there's times when it's trying to

well in terms of heat anyway

um

get rid you know reduce the heat and

sometimes it might want to gain heat

but

it doesn't really care per se it's like

the the niche itself will be have a

greater fitness

and if there's enough fluctuations over

time and enough feedback loops in theory

you'll sort of

gain

the carbon dioxide levels as best can

serve

the overall

production of carbon mass in theory

but in terms of the piece around it's

really it's a good conversation around

um active inference versus the free

energy principle because the free energy

principle or variational free energy

um

jeffrey hinton's work and the

recognition models and work that just

takes you know perceptual

um

data coming in with deep mind can work

with

reducing free energy information but

when you bring action into the game

so that's the interesting thing because

then it's the action that brings in the

energy in a way because

by the action being more efficient

the um

the energy sort of becomes part of the

landscape

so

um whereas the perception
piece
you're kind of tied more to
if there was no actual active influence
and just bayesian influence it's kind of
all information up and down but
once you start bringing action into the
game
that's when it starts to get a bit
more interesting
it's a lot to
think about
information
is usually
framed more on the inference side than
the action side
we know there are information theoretic
approaches to control theory
and there's
all kinds of connections that you know
2021 we can look back at and say that
have been made but
information is
very aligned with knowledge and
inference
whereas
thermodynamics brings in
physical constraints energy and action
like what is temperature if not a
summary descriptor of
the actions of molecules
whereas information
tends to be a little bit more abstracted
and disembodied
and thermodynamics even if it hasn't
been
connected to 4e cognition extended
embedded and cultured etc

it at least is closer to it like the the
pressure volume relationships of a gas
they relate to
its context
and to
how it is actually enacting
physics
so then if we think about in
informational physics
first off how far can we get with
bringing thermodynamics
into information
big topic and then
where is control
and action
in that kind of a model stephen
yeah this is interesting to bring in and
also if you go back to gibbs free energy
and
the way it was traditionally used i mean
you basically have reactants
and you have products which is basically
where we were in terms of an equilibrium
state and where we ended up
and then there's a force that's going to
drive that there's an enthalpic force
i.e an energy change
which will drive things
um
and will succeed depending on the
activation energy of whatever's in
between
which can be calculated
and then there's this kind of
entropic force
which is basically cough cough um
it's just inferred basically by what the
reaction actually did but because you're

only going from equilibrium state to
equilibrium state
whatever happens in the middle was just
like

i don't know

let's be that's basically if you if the
truth be told it's just entropy right so
but that's not it's not just this
black box in this case it's actually all
the stuff that's going on on the
journeys

is where active influence is sort of in
the process so in tropy
or thermodynamic entropy
it's kind of interesting because
there's there is somehow this enthalpic
term in terms of energy that there's an
entropic force they call it in chemistry
which is like how the entropic shift
sort of pushes the reaction
and gives it a force which then combines
with the enthalpic force which is
basically just whether it's going to
give off loads of heat or
absorb heat

oh sorry i should say energy absorbable
um

you know

or emit energy so

this is where the the the

i think it's not entirely clear quite
how that might mix up in and maybe
sometimes it's safe is to not make the
analogy because it's non-equilibrium
and non-equilibrium chemistry has only
really started to happen in any sort of
coherent way since the 1990s in terms of
research it's all been equilibrium state

equilibrium state and

hand-waving in the middle

thanks steven blue

so

i um

like whatever happens in the middle uh i

only had abbreviated p chem and like

this kind of makes me wish that i had

more and you know then like a full

semester of thermodynamics to to really

like be able to screw these pieces

together but but maybe they don't all

actually screw together

with a perfect fit so i mean i think

about okay is it an exothermic or an

endothermic reaction does energy go in

or does energy come out

and essentially all life is

like

all of life is endothermic like some

energy has to go in and where is that

energy in life like because we exist in

this non-equilibrium point like life

isn't

spontaneous and so where is this like

storage happening

and like in some

like maybe it's a very abstract link but

but i think about the energy that going

that goes into creating life as

information because it's essentially

this dna blueprint that we all have that

carries on from generation to generation

that is

the catalyst for the reactions that

occur that create

life

um so i that's just always i don't know

how i think about it
here's here's one image that you'll see
on your
shared screen
so this relates to gibbs free energy i
mean
you mentioned it we've been talking a
little bit about chemistry
and so
this is already pre-adapted
to instead of be the gibbs free energy
it's like
the pain axis so we can
quibble with the specific metaphor but
the the piece that i think is
interesting is the difference between
the thermodynamically favored and the
kinetically favored
reactions
so the green line is kinetically favored
like
if there's a ball rolling it's easier to
climb the b hill
so it's easier to get on that there's
less friction less of a activation
energy cost
and therefore it's kinetically more
accessible to take the green route to
get you to product d
and then that a minus d is the delta
free energy
whereas the thermodynamically favored
product which is like a to e this is
like a complete combustion versus a
partial combustion more energy is
released there's a bigger delta g
but
that product may not be as kinetically

accessible and so this is like classic
ochem
chorus questions
some reactions you want to do them
really hot
because you just want to get over a
kinetic barrier and wherever you end up
is going to be okay
other times you want to carry out the
reaction very slowly
so that you can be sure that the only
high energy balls that come across
are only going to take this green road
and so it doesn't like combust so you
keep the reaction cold
and then in this example
it's already been connected to sort of
behavioral economics like making a bad
decision versus making a good decision
you know you're choosing what to eat for
dinner is it going to be something with
a low
barrier
high kinetic accessibility but maybe
not healthy
but where do we
transmute
from the kinetic versus thermodynamics
of chemicals
to kinetics and thermodynamics of
information and decision-making and
action
and then
the closing the triangle would be like
is there a energetic or a metabolic
account
for decision making like it literally
uses more glucose in your brain to make

taxing decisions
and that would sort of close the loop in
an interesting way
so stephen then dave
yeah this is
so this is also quite useful diagram
because
i mean you'd have that kind of
activation transition state as being the
kind of the hill right so the higher the
hill the more
uh
energy is needed to make that transition
state become available but it is a case
of it going from left to right within
reason i mean there is that idea that
things can move both ways
but what's interesting is
the more dilly-dallian
to use this i know that's an english
phrase um which i do a lot of but
um
the more dilly-dallian is not
necessarily the better thing there right
because the more dilly dally and the
molecules are doing the less they're
going to get over the hump
because it's really a case of this
energy
dynamic and
entropy being
maybe going to help um because
because it in certain cases but normally
it's a hindrance so something to slow it
down but the difference is when you're
dealing with variations
see when you've got variational free
energy

now you can get information the dilly
dallian
gives you
variation in entropy
and that can give you
like entropy itself doesn't necessarily
we don't entry is a funny term but noise
alone doesn't give you any information
variations in noise can give you
information
and variations in noise need a reference
of none well they don't i don't know if
they need that but let's say they it
helps have a non-equilibrium steady
state to reference against
that whole non-equilibrium piece and
complex
um attractor states and all of that sort
of stuff doesn't really feature it's not
an attractor state at the top per se
it's a it's
trans a transition state that
some molecules go to or some
configurations happen to by chance
arrive at so
um
but obviously it's maybe not like that
alone there's this kind of inference
process
yes so this reaction would be reversible
but imagine if the ball gets to here now
it's a 50 50 it could roll either way
it's at the transition point that's what
enzymes stabilize transition state
complexes they reduce activation energy
barriers but now if you were down here
in p2
it would be a positive ΔG

so it'd be energetically uphill because
this is what they call a
thermodynamically irreversible reaction
so K_{eq} like the coefficient of how
complete the reaction is going to
proceed is going to be way to the right
on this one

but if it just looked like a hump
then

in the long run

you'd get

an even number on both sides it's kind
of like ergodicity and so it is there's
a lot that can be seen here so dave and
then blue

yeah the uh advertising meme for that
expensive can opener that needs to be
pushed

that would uh bind to

the

the hump

and actually like use i think you said
flatten the hump effectively

um

[Music]

and how does that happen is there i i
presumably you're getting a reduction of
cognitive dissonance it no longer looks
dissonant to imagine oh spending all
that money i feel so guilty about
buying a forty dollar book the other day
how can i justify buying a twelve dollar
can opener instead of a two dollar one
and the point of sale oh just tap it's
lower kinetic friction versus you know
pull out your um

your shells count them

blue and then stephen

so one critical thing when you're talking about the whether or not a reaction is reversible or not so you when you have the free energy of the product and the reactant if the free energy is the same it's completely a reversible reaction but the fact that it's an irreversible reaction is driven by the lower free energy of the product of the reaction and so

i think about things like um you know even just the self-assembly so we can talk about self-assembly of life of cells but also self-assembly of molecules right so so molecules will spontaneously like you know salt small spontaneously dissolves in water and that's driven by charge right and hydrogen bonds and you know the polar polarity of water and so when you have this um

you know kind of spontaneous reactions that occur that are driven by like molecular charge is that that's again like an information catalyst because the charge is information is it positive or negative is it a north pole is it a south pole and and so that without that kind of information would these reactions

push forward and one way that computers over the last hundred years have revolutionized this discussion is like if you just say well of course information has nothing to do with it we're talking about just chemicals

tossing around in space and we're talking about thermodynamics there's no need to bring information into it and then someone says well i'm talking about the chemical components on my hard drive or i'm talking about the chemistry of this processor

so it's like

there is an information thermodynamic link we are putting energy into this processor or hard drive there is chemistry happening

who will link it

it's no longer a tenable position to say that there just is no linkage or it wouldn't even be useful at the very least it would be extremely useful if not beyond useful to understand this nexus so it's kind of like that info thermodynamic

is already well in progress and we're using devices that are empirical proofs even if the conceptual details are not quite filled in yet blue and then stephen

yeah just to respond to that it's like maxwell's demon

is essentially can be

replaced by a computer right like so a computer can sit on top of the box and decide whether the you know big molecules go this way or the hot molecules go this way and the cold molecules go that way but then you'd have to put the computer in a fridge and you'd have to plug the fridge in right so there's all this like the thought experiment of um you know the

information separation or the
information filtering taking place um
with computers and so that's also like
another direct link between
the info dynamic and thermodynamic um
you know
capacities

thanks steven

yeah this is really interesting um
yeah the this link so what gets really
interesting is when it that when there's
those endothermic so for instance
um actually we used to use this when we
used to do reactions

in the lab so like if you want to you
get a load of ice basically dump a load
of salt into it and then your ice goes
down to like minus 18 degrees
right so now you can start doing
reactions at minus 18 degrees in your
beaker because you can immerse it in
this so

your um

well what's just happened there right
you've just dropped the temperature that
doesn't make sense so this is where this
entropic

you know the ice

the whole way the ice melts and
thaws and

if you know if you drop the temperature
very very

drop the temperature drop the beat
drop the video chat

rip steven

we'll continue though he'll rejoin um
this

slide which i just put the image up this

is like in the wikipedia article on
kinetic versus thermodynamic control
like
one doesn't have to be a chemist
to see that we're taking a few
very
interesting esoteric symmetrical
molecules
and there's some transition state
complex
and then
in the blue if it's very cold
it's going to form this product if it's
very warm it forms a different product
so
the ratio and the temperature and the
solvent there's so many details to
explore
but it's just an example about like
thinking fast and slow type one and type
two conomon you know
decision making frameworks
what's the type one decision
where is it really being thought through
you know hey chill out
cool down slow your horses make the
right decision
versus
um
buying under pressure time pressure for
decisions
that's one of the major threats to
cognitive security and good decision
making is not having enough time
or spending too much time on a decision
which is implicitly taking away time
from another decision because in the lab
we can just heat things up cool them

down but you know we only get one
sequence of action

so

in a broader sense we can't just perform
well

on single decisions because we also need
to do that sort of bayesian
estimation of how much time should we be
spending on this decision blue

so just thinking about time i mean time
is always the

fourth axis right time is always the
other other thing but i mean we have
these things that we say like time is
money right like so so what is the cost
of time

um and like i think about things like
time effort energy and money as all like
things that i put in to drive a reaction
forward like things that i put in to
create a result right time effort energy
and money and so is there some kind of
um

like quantifiable

interchange between time and money or
time and energy or time and effort or
um

i mean it's time always the entropy
increasing thing but like
so okay entropy increases with time
right

but

like i put time into something right
so is this um am i putting entropy into
something to create a result that seems
very backwards

um it and it's just cool to think about
time as like you know something that you

put in to drive a reaction
forward i don't know
and
quality time like attention maybe it's
like you know if we had a unit an erg or
a calorie of attention
you know the the the calories the
temperature heat of some metal sphere
and a cloud you know there's different
amounts of specific heat
and so
maybe individuals attention will be
different it's different to get the
attention of
peer reviewer one versus peer reviewer
two but
is there a commonality for attention and
does that help make decisions
get pushed along the road
blue maybe you could introduce this
nice set of slides on
life-mind continuity
sure um so there's a uh
maybe maybe the first question first and
then we can maybe talk about the
life-mind continuity thesis so
um this is a quote from the paper it
says adaptive active inference requires
that a system autonomously engage in
active inference
maximizing sensory evidence for its own
existence
so
this kind of is a different um yeah and
they cited this reference and this is
from the sims paper
but um
i just thought about

what is evidence for your own existence
like when i pulled this quote out
um
and does that require
self-awareness like like
do i exist i think therefore i am it's
very like descartes and um like that's
evidence for your existence but like
does the little cell
it self-aware enough to know that it
exists i mean it is like the desire to
not be extinguished awareness of your
own existence
um so i think about you know it's
sensory evidence for your model or for
the
validity of your model so that's kind of
always how i think about it model
evidence is not evidence for my
existence it's evidence for the validity
of my model so that's kind of how i've
always thought about it but it just was
interesting this um this phrasing that
was used by by sims uh um because i just
was curious is there a difference
between
evidence for your existence
and
model evidence
and just what do you guys think about
that or or am i the only one that was
like that's kind of weird
there's a lot of
pieces here so one
piece is that in active inference the
preferences are over sensory outcomes
not over hidden state estimates
so

instead of maximizing
um you know values of hidden state
estimates
like making it maximally safe out there
that situation of bodily safety would be
reframed as maximizing evidence for
observing our observations in their
preferred states so one little
flourish that is active inference
germain
is this focus on the sensory evidence
rather than inferred hidden states
um
and then
that is an interesting question about is
it is it the model of itself what is
this how are we getting sensory evidence
for
ourselves and is that the imperative
for action and it just reminded me
because you said um you know i think
therefore i am there this is um
uh latin post hoc ergo propter hook
after this therefore because of this
it's like the post hoc fallacy so i
think therefore i am
and then of course you know cue the 50
fristen papers
i am therefore i think
every possible inversion
like is it that
thought begets
life or does life be get thought
and then if we're only going to find
both of them inextricably tied up
in actual systems
what is the
partitioning or the compartmentalizing

that is

variously meaningful or true or

so that inversion's really cool and and

also um speaks to the podcast that was

released yesterday on the podcast

episode where um it's clipped from i

think live stream five six point no 5.2

so

there's a lot of discussion about change

your mind change your life and then the

inversion of that is change your life

change your mind right i i think

therefore i am i am therefore i think

it's just that's all this action

perception loop of active inference

um

yeah here's the first in paper i am

therefore i think he he did go there

he's been there

this is an interesting um

an interesting

uh paper it it even begins um the first

sentences how does schrodinger's

question which is what is life

touch on philosophical propositions such

as descartes famous proposition cognito

ergosum i think therefore i am

and then a little not exactly a formal

connection but um this weekend is

kognitio 2021

so

kind of another illusion there

but yeah what does it mean

for sensory evidence of our own

existence

is that our um

our proprioception and our our uh thermo

perception and taste

or is this referring to a higher level
of mental activity like as we saw in
the san fed smith paper on mental
action the observables
of one level
so like the outcomes of the higher order
mental model
can be plugged into the priors of a
lower model or the hidden state
estimates of a lower model
so like is observing
our attention
maximizing evidence
for a model which we
pay attention to different things
uh where does where does that play in
with life mind continuity what is that
question
oh
gotta hit unmute i got it
so um the
the distinction that sims made in the
paper between
adaptive active inference and mere
active inference so my question is
whether this distinction
follows the kind of evolutionary late
comer path or if or if it's in line with
the life-mind continuity
thesis and so we can kind of unpack all
of those in the next couple of slides
okay
so the um
life-mind continuity thesis so i i i
pulled from
this paper by kirkoff and froze for what
you see
um

where they asked the question are mental phenomena restricted to living systems and there's several viewpoints here um that the the first one is um obviously you can answer no and there's a couple of different um ways and and so the cognitivist viewpoint uh is that some cognitive systems can be living systems but mines are really associated with computational processes that have semantic contentful properties and minds can be realized independently from life given the right kind of artificial support system and so this this allows um mind to exist artificially essentially uh and then there's the overly generous non-cognitivist free energy principle um thought that that all systems can maintain their variables within a limited range of values all systems that do this can be understood as having some kind of mentality or proto mentality given that the fep casts any system that maintains structural integrity in a fluctuating environment um as engaged in predicting its own future states and this is kind of the pan psychic take on you know that life can exist in or that not that life but the mind can exist

in things like molecules that are
undergoing reactions so there's i mean
in everything there's a mind
thanks for this really helpful breakdown
so
huge topic
you know a lot of implications is it
okay just to throw out a cell phone that
we're done using or would that be
you know a a rights issue
so are mental phenomena one-to-one
restricted necessary sufficient
how is cognition
related to
life as we thermodynamically know it
and some some strong shots from the no
side which is saying um
which first off is always a somewhat
easier position to defend because you
can just say look it's a venn diagram
overlap
and we don't know
so who's to say that it's restricted
because maybe there's some overlap maybe
there's a but not b maybe there's b but
not a so first it's a general claim that
one can deny a restriction
by just saying we're not sure or it
might not always be that way and then
this is just really it's it's clear and
funny how you've written it overly
generous like relative to the
expectations of who maybe someone's very
stingy with their fep interpretation
maybe someone else especially the
authors how they wrote it so
you're confident ferguson that's how
that was their their their um

terms

nice so maybe there's an fep whale out

there for whom this is still not

generous enough but um i i i if it was

cool i didn't mean to to edit it um but

like

there's uh the integrated information

theory people

their

flavor of pan psychism would be like

because all systems are on this

continuum of integrating information

pan psychism

another approach is sort of this

cybernetic pan psychism which is like

because all systems are on the continuum

of

maintaining their variables within a

limited range of values

kind of like good regulator theorem

in other words not because there's an

information continuum but because

there's a cybernetic control

continuum

dot dot

pan psychism

so there's an information centered

inference pan inference equals pan

psychism and here's like a pan control

or pan

resistance to dissipation equals pan

psychism claim

so that's pretty cool how that that

action and inference thread

come back

um in different guises welcome back

steven your surroundings are

less

blue
but i hope that it will work
yep we're on 11 we just
heard some of the no
uh voices
on the question of mental phenomena
restricted to living systems and then
blue
who
is on the other side of the table
well and then there's a couple of
different um views that the authors put
forward in the paper as well on the yes
side
so there is the non-cognitivist
fep so that's the the second view from
the no
but plus um
the
evolutionary late come review so this is
mind but not life requires sophisticated
generated generative neural machinery
that is not present in simple forms of
life such as single-celled organisms and
that was my question about adaptive
active inference versus mirror active
inference is that like putting mind
above
a certain is that at a certain
evolutionary point such as like higher
level vertebrates or whatever only
perform adaptive active inference if
they're evolutionarily at a certain
point or
in a certain as sims kind of implied
in a certain amount of
collectivity like so so uh in his paper
he said that you know

whole organisms can be
can do adaptive active inference but not
like cells within an organism and so
that was kind of
it's along the lines while it doesn't
necessarily um
link directly to sophisticated neural
machinery it seemed like
sophisticated
collective autonomy or like like what
what was the the sophistication and and
where was this line and i kind of pushed
matt on that when he was here but i just
wonder is that
is that linked somehow to this way
um and then i don't know stephen did you
want to say something or and then i can
talk about the life minecon continuity
views yes
yeah um i think actually it can just
kind of continues what you're saying
continues a little bit with the graph
that we had earlier there in the sense
that
i was thinking
when when um
something goes over that that graph and
goes down the other side um
with active inference the like with
normal reactions right you do it anytime
with the same reactants it's the same
basic curve every time right because
it's it's a standard thing but with
life it gets a chance to
know what the final outcomes have been
and start to maybe change the energy of
the hump it can actually change
particularly maybe as well if there's

non-equilibrium steady states in there

but

there's an actual ability to infer

backwards and i suppose that that

inability to infer backwards well in

real time which in real time could be

chemical time

that that could kind of happen through

enzymes or maybe through the mediation

enzymes and whether that's cognition or

whether that's

could be done through some real-time

um you know chemical inference

process um and then of course you've got

that happening over longer times um

and this then becomes when do you start

to move from

a variational free energy into an

expected free energy you know

um but that ability to go backwards

or forwards and start to sculpt

the landscape

of intermediaries

um yeah i wonder i think that kind of

ties into what you're saying

i'm gonna do a little life-mind venn

diagram

what kind of a universe

or world or

simulation or whatever

do life and minds

look like

partial overlap

like in this world let's make this a

little different color

you know in this world

there's things that are having mind but

not life

and vice versa
so
that
is a strict
um
refutation of
the restriction
claim
is it restricted to living systems
this would be no
but then let's just say it is restricted
to living systems
well it could be like that there are
still living systems with no mind
or
could they be the same so
um
how would we know
what would be the observation
or the experiments
or the measurement to make
especially
keeping in
mind or keeping in life that
mind is a little bit of a
potentially
unobservable phenomena so how do we
integrate
sort of
meat and potatoes
with minds
in a way that's going to lead to
testable predictions and optimally
informative experiments blue
well and so that is the the life-mind
continuity visa
thesis is that um there's it's 100
overlap

so the
concept of a mind is grounded in the
concept of life in terms of autocoesis
and
adaptivity but
the problem that i have with that
because the whole reason i brought this
up is like
back to schrodinger's questions what is
life right and so um
it's not
binary necessary necessarily like
there's
some continuum between
life and
not life there's a lot of gray area
there's
you know obligate parasites there's
viruses there's a lot of things that
fall into that like
fuzzy area and then also like going into
the origins of life theory
somehow
some self-replication and autophysis
happened in
non-life that like
drove the transition to life at some
point in our evolutionary history and
and so
where's the mind there
and that's one um other reason that pan
psychists implicitly or explicitly
promote their viewpoints is it there's
no need for like a biogenesis for a
mento genesis for the emergence of mind
from non-mind you just say nope my
belief
is it's all continuums of mind so we're

just like adding grains of sand to a
pile there's a debate about when it's a
heap versus when it's not a heap but
there's no
qualitative
challenge
with explaining the emergence of
complex minds
as described in gerda escherbach
whereas other people who are not pancyst
implicitly it's like saying you do need
to
have an account for how minds if you
think they exist
arise from non-minds or you can take the
anti-psycho position and just say
there's never emergence of mind
so not in the little not in the big and
so there's a few ways to be consistent
you know pan psychism and then
elaboration of mind
uh
mental denialism
which is its own answer or
some sort of qualified intermediate
where you think that it emerges at a
developmental stage or some evolutionary
development or an ecological context
or some informational requirement you
have to be this many bits to have
awareness so there's a lot of
views in the space but this is just
total
fundamentals what systems are living
which ones have mind which ones deserve
to be treated using
respect
steven

this also brings back the multi-scale
question

um

maybe not necessarily i don't know if
physical scale but

temporal scale

you know if something is

actively

variationally

um

enabling itself to um maintain its

steady state

um but it's purely doing it only at the
level of

maintaining a chemical gradient and
that's

that's the whole game

i don't know if that's a bit it depends

when it's not exactly clear where mere

active influence active inference sits

right so there is a slight ontological

question there but the if something's

maintaining itself but it's just doing

one thing and it can do that maybe

through chemical gradients or whatever

but then you've got this temporal depth
of

okay doing that because it's inferring

that by doing that it will allow it to

do something else

at what point do these kind of chain of
events start to become

more life like and

does then then need to be

is that in itself life and does there

almost need to be

a lifelike

entity

that is managing that and then that
becomes mind
and then this is this is that question
that i suppose
fungi raise a bit as well because they
they mediate messaging part message
passing and
a lot of interesting stuff now are they
kind of a mind or not uh i don't know
so

um we could definitely go to talking a
little more about um

[Music]

mirror and adaptive active inference but
it makes me think about you mentioned a
chemical gradient well what are neurons
and glia doing
if not shuttling around
chemical gradients and so it's like okay
is mind
in the sodium ion
okay you know yes or no garden of
forking paths is it in the system of the
lipid membrane and the
transport protein and the gradients it's
like nope that still seems like a fairly
no mind needed physical system okay well
now we're gonna put some other bits of
protein
around it and now we're gonna
converge towards biosemiotics with
signaling at the membrane but it's still
a purely physical system and so it's
like
you start building up
and is there anything left to explain is
that reductionism to just sort of make
everything

literally reduced to the explanation of
the chemicals

and it reminds me of a of a sapolsky
article in 2004 where he says if if free
will is lurking in the
in the uh interstices of the brain then
the crawl spaces are getting smaller and
smaller

because as we get tools that help us
look closer and closer
we're finding there's less and less
wiggle room degrees of freedom

[Music]

tilly lally or whatever you
called it less and less of that
potentially
with deeper and deeper investigation and
dissection but are we just
tearing

something apart not understanding how
it's actually composed blue
so that just reminds me of back going
back to the live stream that um we had
with mike levin

it's like

if we are just
at any given point
our model essentially like dictates our
action right so in under active
inference we have the degenerative model
the recognition model and constantly
these are updated based on what's
happening

in
our world

can we in fact
or do we have a choice when presented
with any given

circumstances in life in our environment
do we have a choice whether or not to
steal a candy bar from its tour or
like like random things that you think
about um
and
this also goes to change your life and
change your mind so it's only through
like practicing piano do we learn to
play piano but do i have any choice but
to sit down and practice the piano given
my existing model or and so is there
like at what point where does free will
come in like
right like does it play in and um mike's
answer was awesome i think he said like
no there's no free will but then he said
okay there's some minuscule
amount of
free will that's non-zero but like the
smallest possible
non-zero number that you could have and
so incrementally then you're able to
effect changes on your model which then
affect the decisions that you make given
your model
so
just two quick things then steven so
with a chaotic system
with a big leopard of exponent so small
changes have non-linear effects
all you need is a tiny tiny bit so let's
not
um confuse the fact that we have small
but potentially even infinitesimal or
fractional free will whatever that means
in the moment with our inability to have
serious consequences through deep time

because it is a chaotic massively
interacting system and the second point
is it's um you mentioned like stealing
something
that 2004 paper with zeki good enough
and sapolsky
is about the frontal cortex in the
criminal justice system and this paper
is really fascinating because it's like
there's two modes to
just i believe centering on the american
legal system but replicated many other
cases either somebody can be like um a
rational sane actor
or they plead a defense which basically
amounts to
being entirely incapacitated so like
either you were fully you should have
known that um the bank robbery would
have gone this way you should have
really waited it out all the ways
or
it was
a situation of which there are multiple
where you cannot be held to be
responsible for the action but we still
need to
you know it's not to get out of jail
free card but it's it is what it is and
so they're
arguing for
a more nuanced perspective
that respects all the different time
scales of changes
that influence our decision making so
it's a really and and you know this is
20 almost 20 years ago so a lot has
changed in neuroscience and in law and

society but this is a little bit of an
older
peep
so stephen then dave
now i think this is quite an interesting
point that you're making here i i i on
that slide if you go back to that slide
that has the mind on it
i'm just trying something here if a
little idea i i've just got it
if
okay you say so if i was to put life you
see there's life there put life over
mind okay
so i'll do actually just do that so it's
like this
okay so say
this is i don't know this is an idea so
say you've got
life as in the organism's body
and then you've got life in terms of the
the the the cognitive
um
neuronal
life
and
they're like they're differentiated into
two types of organism
in the organism
and the two organisms talk to each other
and the boundary between the two talking
to each other is mind as an emergent
process
something that's coming up
as an idea
very
interactionist and relational and
interfacial that maybe we shouldn't be

looking for the reduced essence of

atomic

modular life

but we should um relentlessly pursue it

in the interfaces

dave

yeah the um

if mind is entirely a matter of memes

and memes need material processes to

live in and material processes need

means to keep themselves from falling

apart then things become a lot simpler

uh but i actually want to make a comment

on the insanity plea i think it might

have been dershowitz who went back and

looked at the statistics of successful

and unsuccessful

insanity pleas and he found that when

people when somebody is so completely

berserk that you can't even begin to

empathize with him juries say

he's just crazy kill it

but the places that people the juries by

the the insanity plea is the juror

saying to himself although he never

quite comes out and says this

you know

under the circumstances i could see

killing the guy that's all right

whatever it takes to get him off uh you

call that insanity okay we'll get him

off with insanity

so it's exactly the opposite of what

you'd think you know the supposed

elizabethan argument that well the guy

didn't know the difference between right

and wrong so you can't hold him

responsible american juries don't think

that way

big questions about

what the legal system is for and what it
does flu and then there's a question in
the chat

so just on to like the legal system and
i i wasn't like meaning to start spawn
this entire conversation with the
stealing candy bar

but but you know so much of our
creation of the model of the world
occurs when we're young and in early
development and and i think that um
you know at least in the states the the
amount of un
unacknowledged childhood trauma that
happens um

for poverty and for drug abuse reason i
mean there's a infinite number of
reasons that people undergo some kind of
abuse as a child

all of that like really contributes to
how well you're going to function
in society in later in life and i don't
think like while there there are people
that plead insanity and there are some
cases of of severe abuse and whatever
where the people get off but but
so much so many of the people that are
in our prison systems and go through
juvenile systems have these horrible
early experiences that really are
happening when their model of the world
is being solidified

and so for me it really um
it explains a lot of the choices that
are made um
by people in those circumstances

and it's helpful to think about it in
terms of a trajectory we're not just
um maybe taking a snapshot moment and
just doing a snap judgment but we want a
trajectory of
healing and of um
health
stephen and then dave and then we'll ask
the question
yeah this idea because we talk about
people being out of their mind you know
and in some ways like you say these if
someone has an experience or something
or
some biological
condition the idea is it's like
the
i suppose the biological
architecture
means that the person's ability to
mindfully
choose isn't there in some ways it's
almost like
the yeah either the the somatic
priors are so
skewed or the uh
the mental architecture is so distorted
that
a coherent mind isn't available
thanks steve
i met a lawyer
who worked with the um
criminal justice system in chicago
on um child abuse cases
and she's had a huge number of these
cases of serious abuse like a
three-month-old being put out on a fire
escape in the middle of winter

to toughen him up
and she know instead of just saying oh
this is so horrible you got to put the
mother in prison
they interviewed her says what do you do
with babies and she had no idea she had
never seen even minimally competent
child rear
from anybody
and they found you explain to these
folks you've given you know the
supplemental education to understand why
middle class people treat babies a
certain way they think that's great oh
that makes sense i just didn't know you
could do that i thought you had to beat
them they wouldn't stop crying if you
don't beat them
okay
and say you know you put them on a long
time probation that is not punitive at
all just so you can keep track of them
and keep the education going and keep
them away from the crazies and just it
was working extraordinarily well
thanks so here's a question from the
chat i'll read the question and then
i'll read one response in the chat
already and then um anyone can
jump in and we have some slides that
speak to this so joseph wrote
could the extent of how conscious
something is
be measured on the basis of how far
ahead in time an organism cell creature
can plan or minimize expected free
energy and then
um

i know that some people will probably
have things to say on that but tucker in
the chat wrote
under this idea
then might the entropy chaoticness of
the environment influence the degree of
consciousness of a system a more dynamic
system or environment would require a
more adaptive agent in order to maintain
negantropic coherence through time
the more entropy the need for a better
model
super interesting questions and
responses
so i guess just uh glancing off of that
uh legal and choice discussion into more
broader questions it's so fascinating
how
especially if we're going to be locating
function in the interface between the
agent and the environment
maybe depending on whether you're in um
an unmarked room or a very
interactable space
consciousness is going to be different
are you more or less aware and then the
other piece that comes to mind and then
stephen and anyone else is um
from livestream 25
computational boundary of a self
and um there's the past
and then there's the future below and so
the past cone is kind of like the the
depth and the breadth of what has
influenced the system
and then the uh
the future cone is about the scope of
planning so it's a really

interesting question about temporal
horizon but is it enough to just plan
deeply like if you just plan
sparsely but deeply over a hundred years
versus a very rich
plan

very thick plan for 10 minutes
which one of those is there gonna be a
single currency that describes or
compares those two
how do we think about planning in
heterogeneous systems
that

have different affordances or have
different complexities of their internal
generative model

so there's a lot of great
questions in this area stephen then
anyone
else

yeah these are really good questions um
i suppose if we're looking at measures i
mean there's also this interesting
question of

not necessarily what you're able to do
but how far you're able to get with what
you're not able to do like the
like when someone's really working hard
cognitively it's like
like i mean
you know

if i want to predict what the inside of
a jar is going to be like i can just
boil the liquid and it becomes sterile
and i know it'll be like that for a long
time

um the question becomes when you've got
these intractable kind of

multifaceted things a bit like what's
going to happen to a
a collection of
family members when they all meet
together for thanksgiving right
um this
that that
that that temple depth is is definitely
a good point and i suppose there's that
question of um
how to penetrate
time
when it's hard to penetrate as much as
how much are they able to plan but i'm
sure the two things are linked and you
could do both
blue and then i'll have a thought
uh give your thought first daniel and
because i'm gonna go on a tangent okay i
wanted to connect it to
the paper that we're allegedly
discussing
and this user resource
and mirror and adaptive active inference
model
so on the uni-directional the left side
you have like a more adaptive agent it's
here a spider and then you have this
less adaptive but doing mere
active inference like a web okay so it's
like a person with a thermometer or a
spider with their web
versus this reciprocal case
where on both sides of the table there's
a user
and then
we've heard before about how the time
horizon

of planning relates to the um the depth
or the quality of the conscious
experience for example the 2018 alias
interview with myself and martin and
carl fristen in the final question he
even talks about the temporal scope and
the existence of counterfactuals as
being relevant for generating
consciousness
so then it made me wonder
what if by going reciprocal
multi-integration
you actually shorten the time scale of
planning like if i'm gonna um
set out a buffet for myself it might
take hours of planning
and so there's a lot of um mere active
inference happening on the stovetop but
it's a very
complex process one user and takes hours
of planning but then to
cook with a friends
might be just
a minute of planning we're going to
agree to cook together and so actually
the temporal explicit depth of planning
in an improv improvisational or
interactionist
context
the depth
drops potentially
because you might not need to
explicitly prepare for so many
counterfactuals
and it's not like you're just moving
pieces on the chess board
you just need to take that first step
listen to what they're gonna say

then think about what your response is
so then that almost challenges
the um
simple time horizon or counterfactual
basis
of cognition or phenomenology
because um
and it's a such a helpful distinction
made by matt that i know we're going to
use in other papers and research that
these are really different kinds of
systems this left system is stigmatic
and it's still a collective system with
emergent properties but then this kind
of double stigma g interactionism
brings us into a new territory that's
very important to delineate
stephen
yeah this and this takes us into that
realm of sense making different types of
sense making because it's action
yeah being able to
act and interact
in appropriate ways
and in many ways the more appropriate
ways can be stimulated by things in the
environment i mean
the fact that we have a very expressive
face in a way
puts our face out there as something we
can inscribe
and use in that process right we just
look at each other's faces
now
for all the psychology in the world no
one quite know you know it's it's it's
all it's all it's all an interpretation
but the the thing that i think this

really highlights is
the game in town is action policy
it doesn't the game isn't for us to walk
around as humans with big cognitive
capacities on the top of our skull or in
our skull the game is that we can
interact and sense make
um the ability to like you say to plan
on our own
is um
is a nicety
but actually it's
been a bit of a
we could say a wrong turn for the last
200 years because it's all been about
thinking as being the game
and feelings being this kind of artifact
that we've
we try and minimize as some sort of side
product that's somehow you know
a weakness of the human mind or
something and and it's so far from the
truth because it's
pretty much what it was like before we
took that wrong turn it's actually about
how to act and feel in the world so
this is a good point
thanks steven blue
so i'm going to go off on my tangent now
but i really just want to address the
first
question in the chat about how
consciousness relates to
our ability to plan in in how far out
that we plan and and so i'm reminded of
and maybe daniel you can help me
remember if it was
deeply felt affect or sophisticated

active inference

but in in one of those papers we talked

about um

actually

anxiety as trying to plan too far out in

the future right so if you're trying to

plan too many time steps ahead then that

that is like the cause of a

neuroatypical situation

so i'm not sure i will and then i think

that they said that maybe depression was

staying too many steps behind or

thinking about too many uh steps in the

past

so thinking about um

and it was so my takeaway from that was

staying in

the present moment

and so i don't know that the that um

just briefly to respond to steven's

comment i don't know that um having this

big cognitive

um

giant capacity giant cognitive capacity

is for interacting

as much as it is for adapting

um because humans like cockroaches adapt

to you know infinite circumstances right

we live on all or are present on all

continents in the world and and i mean

even antarctica there's some research

scientists there right now i'm sure

digging holes in the glaciers to find

out how much our ocean is going to rise

i hope

that somebody's there doing that anyway

i'm not sure who's funding that research

but but hopefully somebody's doing that

so

i don't know i think it's more adaptive
the cognitive capacity than anything
else

and to so good good throwback there with
the uh sophisticated and deep active
inference and the anxiety the parametric
investigations of anxiety that we've
been on before

and one day when we have better keyword
indexing if anyone is passionate about
this

they should collaborate with us because
how cool would it be

to have deeply

richly

indexed keywords for these papers and
artifacts and discussions and have them
translated

into multiple human languages and
computer languages so and graphics so
that we could be really

clear

um

so i mean

there's a lot to say here and the only
other throwback that i was going to add
in was to way long ago in number 29

active inference

we talked about the scaling relationship
of the ants brains with a larger colony
having a proportionally smaller ant
brain whereas larger mammalian societies
have larger mammalian brains because
mammalian sociality has to do with like
memory of repeated interactions and
theory of mind and the facial
recognition all these things that we're

talking about here whereas potentially
colony organismality
means that
the increased scope
or
capacity of the colony
can occur over evolutionary times
through a diminution of the scope or of
the cognitive processing on board each
ant so joseph's question
it's sort of it's just such a great
question so thanks again for asking it
because
we can sort of look at it at the
i mean it's even asked across scales
but what if like the cell
plans less
because it's it's part of a high trust
system so now the the beta cell in the
in the pancreas doesn't have to have a
deep temporal model its temporal model
just has to be like one step
but it's playing a role in a system that
now has a circadian regulation of sleep
or a blood sugar
and so where
do we ascribe
consciousness or awareness or cognitive
possibility across different systems
which again brings us back to um from 25
these nested cones of cognition and
their different shapes and their
attributes blue
um i just wanted to shout out to tucker
as i was like i don't know drawing this
big thing uh
with the last question so tucker
actually made the link between

consciousness and adaptivity and i kind
of just stole it i stole all the thunder
um but but really uh
and are those two things linked right so
so planning very far in the in the
future versus the having the ability to
respond to whatever comes your way in
one time step those are really different
like one seems very rigid and the other
seems very fluid
and and tucker actually um
kind of used consciousness and
adaptivity interchangeably which is an
interesting
connection and thought um i don't know
maybe resilience and that adaptivity is
is kind of what i would use but but it's
interesting to think about
thanks steven
yeah and also i like that point you're
making and suppose it ties with that
adaptivity about the pancreas um
you know you if you've got the pancreas
gives you the affordance of modulating
stuff in the body as do these other
uh organs but like you say these cells
and
if i have a
crude or
multiple ways of
mindfully or
livingly i don't know if that's a word
adapting that you can do something but
you got no pancreas or you've got no
cells that give you an affordance to
actually change it you can have all the
computational
cognitive modeling whatever it is

you can't do anything it's literally a
conceptual framework without any way to
do that sort of control so you know the
more important thing is to have
something which can
follow through at these across the
scales
and
change how they are maybe cybernetically
integrated
than it is to have the like i say it's
not all about cognition it's about
action policies and i think that's
really interesting
so earlier we had just those three right
just cognition
agency and autonomy
and then that was um
being inspired by just some of the
quotes from the paper where those terms
were used but we're thinking about
that three
then we've talked about like adapting
and being flexible and learning over
uh developmental as well as evolutionary
time skills we can think about the past
and the future like the time cone
illustration to what extent does memory
matter i mean can you have a memoryless
awareness
or is awareness
requiring memory or can you have
awareness without planning
some meditation practices might lead one
to think so
or is planning vital
what about the capacity for interaction
if we don't have interactions

highlighted

you know paging dean

are we falling victim into

a modular

reductionist nightmare

or how do we hold

interactionism at every step

what about

kind of complex systems properties like

resilience or anti-fragility or

hysteresis phase transitions

those are descriptors of various systems

and are those just coming along for free

with these do they grant the ability for

these are they correlated dimensions are

they uncorrelated

and then

the

million dollar question maybe we need to

scale that for inflation you know the 10

bitcoin question

where does consciousness come into play

and is that something we can ever get a

handle on or is consciousness always

going to be at the fringes of our

explanations

and some of these more accessible or

opaque features

will be amenable to quantitative

descriptions whereas this

um might be always in a different

category so blue

well i think that they're quantifying

consciousness already

but um memory is really what i wanted to

talk about here the key word that's on

this list and so

i mean i don't know i think about memory

right like there's the rom and there's

the ram

um

and uh

is memory or to what degree is memory

required for a generative model

and so

i i just wonder like i mean all of the

experiences contribute to the formation

of the model and what is contained in

the model is

effectively like our action course i

mean what's in the generative model does

anyone know um i think about it like

like what is um the actual i think about

it like an action course or like the

response

i don't know i'd love to hear what other

people think it's in the generative

model what's inside of that um and also

um whether or not memory is required

stephen

i think this also you can bring in there

with like affordances coming they say

you know like you say interaction

capacity

with affordances

uh i suppose you could say memory

is an interaction capacity

with the affordances to whole priors you

know they

um so there's an interest in um

there's an interesting sort of

dynamic here between

these processes and

the affordances

and

normally affordances are

well for practical purposes are normally
thought of or used in terms of
something a bit more concrete and
at the human scale but
when it kind of is multi-scale as well
isn't it
you know if we're going to talk about
mental action
and thought as mental planning
then absolutely
there are mental affordances in the
model as formalized by sandved smith and
others
and then to blue's memory question about
where it's related to the generative
model so i thought of a few different
options not exhaustive so the generative
model could have an implicit onboard
memory like that d
prior
that could just be a summary variable
that summarizes memory so you can just
say the past moments temperature here's
my prior on the past moment's
temperature
or
the past sequence of temperatures could
be explicitly represented as like kind
of like a list or a vector so
there could be on-board memory like
within the internal states the internal
generative model of the active inference
agent
could either have just sort of a
snapshot summary implicit memory which
is good enough in many cases so we might
just model it as if anyway or there
could be this explicit

just like a policy is a sequence of
actions you could have a multi-time step
prior over
previous states aka memory but then the
other

[Music]

option

is like this extended or stigmatic
memory with a recognition model like do
i remember what i looked like at that
one family dinner when i was three
no would i recognize the picture
yes would i know where to look for the
picture yes

physically what about asking somebody so
we're thinking okay affordances social
affordances spatiotemporal location of
extended cognitive process recognition
model

but do i remember that
or am i merely remembering or preferring
certain affordances that bring me
to the edge of that abyss
and then the recognition model takes
over once the picture has been
physically presented that's the stigma g
and then

just using matt's distinction

that's there's you in the photo library

that's the unidirectional

multi-scale inference integration

and then there's the conversation where
we actually spur each other's memory
uh and that's where we get that sort of
next level of reciprocal integration to
kind of bring it back to that idea blue
really nice with the extended sigmoidy
recognition model um

and i think that like there's some
like ram some kind of ram is required
for any generative model like like i
think about the generative model a lot
like the software that we're running and
in order to have software you have to
have some ram
um but but
is the long-term
memory required
also like so or is the long-term memory
just what goes into creating
the generative model and then all of
that can be forgotten and and what i'm
thinking about that i'm thinking about
people that have lost their memory for
whatever reason um like so so cases of
amnesia like the person can't remember
anything like i think about um the old
movie memento and like you know every
day i can't remember anything i can't
remember anything but like i still know
how to brush my teeth or i still feel
like when i wake up in the morning i
need to brush my teeth or i need to
shower or i remember what food is right
like so i don't need to like you know
try this or that to see if it's edible
again so there's some
things that are maybe stored in
the generative model and maybe some
things are irrelevant
thanks blue steven
yeah this
this is um ties into
the action policies of
remembering like if all memories is an
act of

imagination

is an act of

recapitulation

then

what needs to be present as you're

saying is is a way to evoke

affordances and you could say that the

affordances of

remapping

uh sensory states you know maybe where

you look with your eyes and maybe when

someone's amnesic it's not that they

haven't got the priors or the knowledge

they haven't got the action

capacity to recapitulate

from

whatever priors were there i mean this

is a bit speculative

but uh

anyway

it reminds me of the uh neuroscience on

taxi drivers and spatial reasoning and

general relationships between spatial

memory and practice there and cognitive

capacity and preventing cognitive

decline and memory palaces like memory

palace is kind of this weird quirk i

mean how weird you can imagine a place

with file cabinets and walk around it

and have stories about that place and

remember you know pi to 5 000 digits or

remember the iliad how weird oh wait

mental action

spatial cognition

narratives of action

it's like

you can build that physical palace and

have the photos there

or

you can update your generative model
to include

a spatial context where those cues exist

and so it's just a very

interactionist way of thinking

about architecture including mental

architectures

like you just need to have the exit sign

and then when somebody wants to leave

they're going to recognize it and walk

towards it and then the next step

will be there and so the architect puts

the next step there safely and

accordingly to function but when we're

the interior mental designer or

architect we can make spaces that also

prompt us

and so it's not like we're storing the

memory itself in that location were

remembering a very compressed version of

the cues that interact with us to then

generate um upon demand

stephen

yeah that's a good example the memory

palace because

is is the

question

that when something's laid down in such

a palace

if you go to visit

the palace and you get this whatever

sensory cues that come up

it's um

it's solid enough that you will come

back up with us with a particular answer

you know what i mean you'll remember x y

and z

from that so it appears like you've
remembered the thing
so to speak but actually
it's just that that combination of
bayesian inferences
will
lead you to act to recapitulate it in
the way
now obviously the average person can't
do it as well as these trained experts
but and they they can actually do it in
a very sophisticated way
but um
because the in terms of pure floating
ideas is that whole they say six plus or
minus one is all you can keep in your
brain at any one time in terms of a
just a concept that's floating and
that's uh but
then that obviously that that isn't
what's the case when you can start to
use memory palaces so it's a very
different
thing going on
it reminds me of um one
class as we go to our closing notes a
class where just each week the homework
and class was to memorize a poem and
then recite it
that's kind of how education has often
proceeded
and it's something that um
is just fundamental for oral cultures
and has been a skill that is less
emphasized in literate cultures
and
there's that dense
cross-linking

of like rhyme and narrative and tonality
and rhythm
so that you really don't need to keep
50 things or 80 lines in your working
memory in your ram with a
computationalist metaphor
what is being remembered or recalled or
prompted
and that investigation well if it's not
being just stored as a text file
what is it
and then it's just so cool how it
dovetails very nicely with deontic cues
and ostensive cues in active inference
and conversational
theory and a lot of other areas
blue and then we'll have last comments
here
so just what you guys we're both talking
about um reminded me of cal newport's
book
deep work if you haven't read it i
highly recommend it it's awesome but in
that book he talks about um creating a
memory palace and learning how to
memorize a deck of cards as an old skill
that people just don't have anymore so
you construct the memory palace you
learn to have a set of cards memorize
instead of like you so you can literally
memorize 52 things in your memory palace
anytime so you can memorize the order of
a deck of cards just like he says to
build this as a skill because it's
enhancing of
memory and i wonder if this isn't going
to be something you know as we're
becoming more and more illiterate um

which is it's an interesting i think
we've reached like the crescendo of
literacy at least in the in the us
um and have we reached this crescendo
because of

technologies like this like i don't need
to read that paper i'm just gonna listen
to this group of people talk about it
right and get the key points out of the
paper and their summary and their
discussion and then i'll have some
interesting things to say about that
paper um and so is it is it art kind of
technology that's paving the way for
illiteracy again and will memory become
increasingly important is great to think
about and i'll just give my final
thought here um thanks for the
interesting discussion today and to matt
also for the for the provocative paper
and i'm looking forward to reading and
discussing more of his work
thanks blue steven or dave any last
comments

yeah i do quickly thank you again a
really good conversation and uh
yeah i think there's definitely
something here

um
around maybe the idea of adjacent
possibles as well so was there something
about the memory palace that one thing
is then an adjacency to the next thing
when you have to hold something on its
own

it's going to have to be supported by
its own infrastructure i suppose
and that's quite an overhead

um

so yeah

because there's also this thing of how long you can hold something in your memory without having a way to ground it it just sort of fizzles maybe for some people quicker than others but uh so yeah all very good i'd have to go back and think over a number of points came up here that we could expand on in the future

thank you stephen dave any last comments

nope just gotta followed up on these references thanks all thank you well

to matt

sims

for writing this great paper and joining our dot one and watching the dot zero for um dave and blue and stephen for joining and for for helping in different ways on the dot zero

that

brings us to the end of

30.2

and next week's

at this same time

we're going to be in 31

non-equilibrium thermodynamics and the free energy principle in biology by patricia palacios and matteo colombo so at least one author will join at least one week

uh

stay in touch

let us know if you want to join for 31

or beyond

and see you all later